

**TABLE 8.1 Great Numbers Are Worth Repeating:
The Probability of Survival at Age 65**

To Age:	Female	Male
70	93.9%	92.2%
75	85.0%	81.3%
80	72.3%	65.9%
85	55.8%	45.5%
90	34.8%	23.7%
95	15.6%	7.7%

Using a “moderate” mortality assumption.
Data Source: RP2000 Mortality table; IFID Centre calculations.

I would definitely have to agree that in reality this would never happen. Certainly, nobody would withdraw that final year’s sum of \$16,000 and then say, “Oops, what do I do next year?” But the idea is to plan ahead and to realize the consequences of your actions in their most drastic, worst-case scenarios. To avoid a potential crisis, you must do one of two things, at this point:

- Reassess your asset allocation to determine whether your total capital allocation would permit you to invest more aggressively so that you earn more return on your nest egg
- Cut down on your consumption—in other words, reduce your needs

But here is the \$200,000 question. What if you don’t know exactly what your rate of return will be? In an ideal world, everyone would know the rate of return for every asset class and the exact rate of inflation during retirement; therefore, you could figure out precisely when your money would run out.

So how can we perform this exercise in the real world, where market returns fluctuate on a daily basis and the length of a human lifespan is so uncertain?

Well, here is where a different set of probabilities come in. We are now ready for the full-fledged model. I call it the “Dual Uncertainty Model” because two sources of uncertainty must be dealt with here: future investment returns and mortality rates.